Maastricht University

Department of Advanced Computing Sciences

MATSE: Operations Research

Lecture 9. Game Theory

[based on slides and notes by: Frank Thuijsman, Steven Chaplick]

Two-Player, Zero-Sum Games

Setup: Two players Alice and Bob act **simultaneously**
(Alice uses strategy a_i; Bob uses b_j)
Alice's score is s_{ij} and Bob's is $-s_{ij}$

Alice's payoff matrix

$$\begin{array}{c|c} \text{A plays} \backslash \text{B plays} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \end{array}$$

Bob's payoff matrix

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Called **Zero-Sum** because each play (a_i, b_j) leads to a total score of **zero**.

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To analyze such games, we must assume *rational* players.

- If we know Alice's play, then Bob can plan a "best" counter strategy.
- But, players act simultaneously, maybe one strategy is more likely than another?

  $\rightsquigarrow$ Analyze such games by probabilistic play!

Two-Player, Zero-Sum

Some games have redundant options , or a deterministic strategy.

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- Maybe some rows (cols.) would never be played ...

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B plays

A plays

$$\begin{array}{c c} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 3 & 1 \end{bmatrix} \end{array}$$

Would Alice every play the new row a_3 ?

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$\rightsquigarrow$ a_3 is dominated by a_2.

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Thus, always better to use a_2 than a_3 !

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What if Alice plays a_1?

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But Bob playing b_1 $\rightsquigarrow$ Alice playing a_2

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But Bob playing $b_1 \rightsquigarrow$ Alice playing a_2
But Bob playing $b_2 \rightsquigarrow$ Alice playing a_1

unstable! :(

Saddle points

When does a game have a **pure** strategy, i.e., where Alice picks a_i and Bob picks b_j?

$$\begin{array}{c} \scriptstyle{\text{A plays}} \backslash \scriptstyle{\text{B plays}} \\ \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\ \begin{bmatrix} 1 & 3 & 0 \\ 4 & 2 & -2 \\ 3 & 1 & -1 \end{bmatrix} \end{array}$$

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- (i, j) is a *saddle point*, when S_{ij} is both the min entry in row i and the max entry in col j.

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stable!

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- (i, j) is a *saddle point*, when S_{ij} is both the min entry in row i and the max entry in col j.

$\rightsquigarrow$ A saddle point (i, j) provides an *optimal* solution, the *value* being S_{ij} for Alice and $-S_{ij}$ for Bob.

Multiple Saddle Points

What if there are multiple saddle points?

Suppose that an $(n \times m)$ matrix S has two saddle points (i, j) and (i', j'). Show that (a) $S_{ij} = S_{i'j'}$, and (b) (i', j) and (i, j') are also saddle points.

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Consider (i', j) and (i, j'):

- $S_{i'j}$ is in the same column j of S_{ij}

  $\Rightarrow$ $S_{i'j} \leq S_{ij}$ because (i, j) is a saddle point
  It is the max entry in the column

- $S_{i'j}$ is in the same row i' of $S_{i'j'}$

  $\Rightarrow$ $S_{i'j} \geq S_{i'j'}$ because (i', j') is a saddle point
  It is the min entry in the row

- $S_{ij'}$ is in the same column j' of $S_{i'j'}$

  $\Rightarrow$ $S_{ij'} \leq S_{i'j'}$

- $S_{ij'}$ is in the same row i of $S_{ij'}$

  $\Rightarrow$ $S_{ij'} \geq S_{ij}$

$$\begin{bmatrix} S_{ij} & S_{ij'} \\ S_{i'j} & S_{i'j'} \end{bmatrix}$$

Two-Player, Zero-Sum Games

Probabilistic play.

Bob's loss matrix

B plays

A plays

Alice's payoff matrix

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Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p, and a_1 with prob. $1-p$.

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Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p, and a_1 with prob. $1 - p$.
- What p should Alice pick?

Bob's loss matrix

Alice's payoff matrix

B plays / A plays

$$\begin{array}{cc}
 & \begin{array}{cc} b_1 & b_2 \end{array} \\
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Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p, and a_1 with prob. $1 - p$.
- What p should Alice pick?
  best guaranteed payoff

Bob's loss matrix

Alice's payoff matrix

$$
\begin{array}{c|c}
 & \text{B plays} \\
\text{A plays} & \begin{array}{cc} b_1 & b_2 \end{array} \\
\begin{array}{c} a_1 \\ a_2 \end{array} & \left[\begin{array}{cc} 1 & 3 \\ 4 & 2 \end{array}\right] \begin{array}{c} 1-p \\ p \end{array}
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Two-Player, Zero-Sum Games

Probabilistic play.

- Alice plays a_2 with prob. p, and a_1 with prob. $1-p$.
- What p should Alice pick?

  best guaranteed payoff

  pick p to max the min payoff.

Bob's loss matrix

Alice's payoff matrix

$$
\begin{array}{c|cc|c}
\text{A plays} \backslash \text{B plays} & b_1 & b_2 & \\
\hline
a_1 & 1 & 3 & 1-p \\
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Two-Player, Zero-Sum Games

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Bob's loss matrix

Alice's payoff matrix

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$\Rightarrow$ Expected Payoff

$\Rightarrow$ Graphical Method

Two-Player, Zero-Sum Games

Bob's loss matrix

Alice's payoff matrix

$$\begin{array}{c c} & \begin{array}{cc} \frac{1}{4} & \frac{3}{4} \end{array} \\ \begin{array}{c} \frac{1}{2} \\ \frac{1}{2} \end{array} & \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \end{array}$$

A plays / B plays

Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are *optimal* since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

The game's *value* is this max min payoff for the first player, here 2.5.

Two-Player, Zero-Sum Games

Bob’s loss matrix

Alice’s payoff matrix

B plays

A plays

$$\begin{array}{c|cc} & \frac{1}{4} & \frac{3}{4} \\ \hline \frac{1}{2} & 1 & 3 \\ \frac{1}{2} & 4 & 2 \end{array}$$

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Strategies $(\frac{1}{2}, \frac{1}{2})$ for A, $(\frac{1}{4}, \frac{3}{4})$ for B are *optimal* since they both ensure to **maximize the minimum payoff** and (symmetrically) both **minimize the maximum loss**.

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John von Neumann

[1928] The **MiniMax Theorem**

Every $(n \times m)$ matrix S has a *value*, i.e., there is a unique number v with

$$\max_x \min_y xSy^T = v = \min_y \max_x xSy^T$$

row vectors $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$

A bigger example

$$\begin{array}{c c}
{\scriptstyle \text{B plays}} \backslash {\scriptstyle \text{A plays}} & \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\
\begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \\ 3 & 2 & -3 \end{bmatrix}
\end{array}$$

A bigger example

$$\begin{array}{c} \text{B plays} \\ \text{A plays} \end{array}$$

$$\begin{array}{c c} & \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \\ 3 & 2 & -3 \end{bmatrix} \end{array}$$

- Check for dominated rows/cols

A bigger example

B plays

A plays

$$\begin{array}{c|ccc} & b_1 & b_2 & b_3 \\ \hline a_1 & 0 & -2 & 5 \\ a_2 & 4 & 10 & -1 \\ \cancel{a_3} & \cancel{3} & \cancel{2} & \cancel{-3} \end{array}$$

- Check for dominated rows/cols

A bigger example

B plays

A plays

$$\begin{array}{c c} & \begin{array}{ccc} b_1 & b_2 & b_3 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} \end{array}$$

- Check for dominated rows/cols

A bigger example

$$\begin{array}{c|ccc} \text{A plays} \backslash \text{B plays} & b_1 & b_2 & b_3 \\ \hline a_1 & 0 & -2 & 5 \\ a_2 & 4 & 10 & -1 \end{array}$$

- Check for dominated rows/cols

  any more?

p

A bigger example

$$\begin{array}{c|ccc} \text{A plays} \backslash \text{B plays} & b_1 & b_2 & b_3 \\ \hline a_1 & 0 & -2 & 5 \\ a_2 & 4 & 10 & -1 \end{array}$$

- Check for dominated rows/cols
  any more? No!
- Check for saddle points

p

A bigger example

$$\begin{array}{c|ccc} \text{A plays} \backslash \text{B plays} & b_1 & b_2 & b_3 \\ \hline a_1 & 0 & -2 & 5 \\ a_2 & 4 & 10 & -1 \end{array}$$

- Check for dominated rows/cols
  any more? No!
- Check for saddle points
  no saddle points $\rightsquigarrow$ consider a mixed strategy

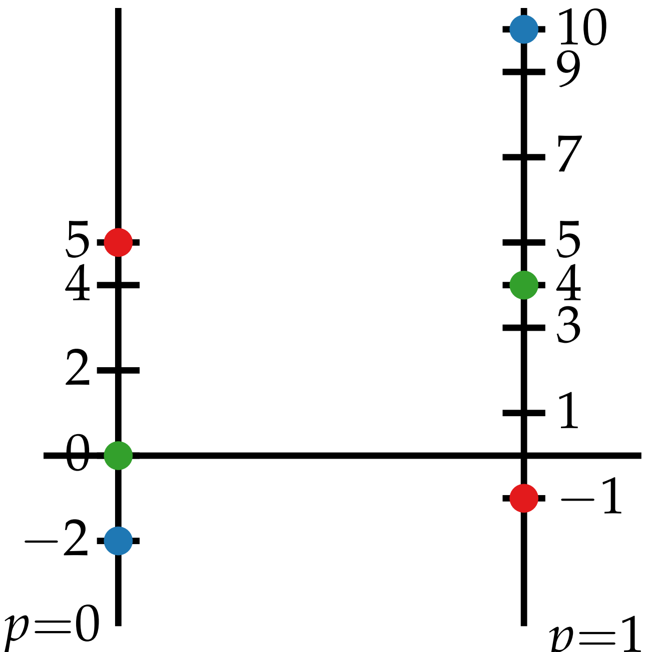
p

A bigger example

$$
\begin{array}{ccc}
 & & \text{B plays} \\
\text{A plays} & & \begin{array}{ccc} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{array} \\
\begin{array}{c} 1-p \\ p \end{array} & \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix}
\end{array}
$$

A bigger example

$$\begin{array}{cc} & & \color{blue}{q_1} & \color{blue}{q_2} & \color{blue}{q_3} \\ & \text{B plays} \backslash \text{A plays} & b_1 & b_2 & b_3 \\ \color{blue}{1-p} & a_1 & & & \\ \color{blue}{p} & a_2 & & & \end{array} \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix}$$

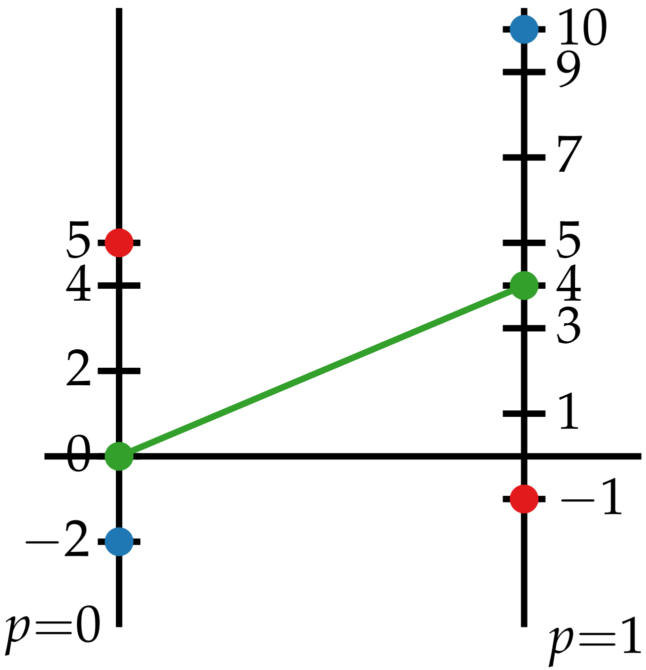


A bigger example

B plays

A plays

$$\begin{array}{cc|ccc} & & q_1 & q_2 & q_3 \\ & & b_1 & b_2 & b_3 \\ 1-p & a_1 & \boxed{0} & -2 & 5 \\ p & a_2 & \boxed{4} & 10 & -1 \\ & & 4p & & \end{array}$$

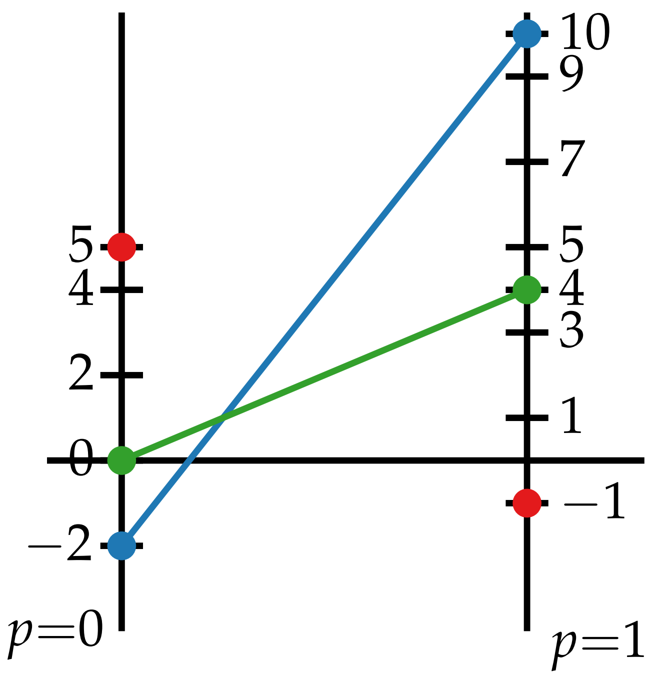


A bigger example

B plays

A plays

$$\begin{array}{cc} & \begin{array}{ccc} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{array} \\ \begin{array}{cc} 1-p & a_1 \\ p & a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} \\ & \begin{array}{ccc} 4p & 12p-2 & \end{array} \end{array}$$

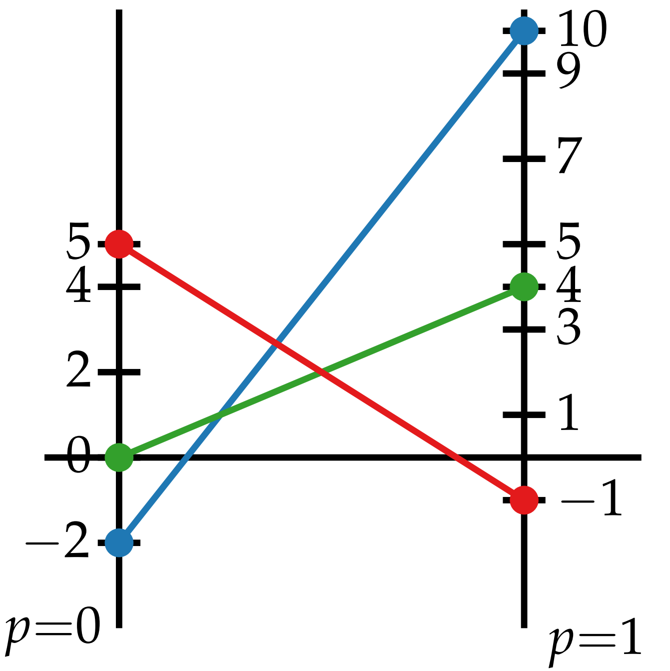


A bigger example

B plays

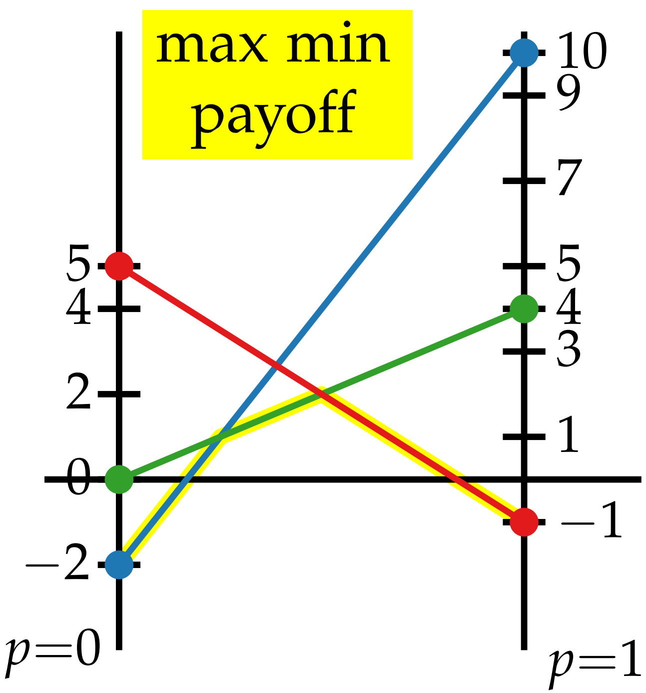
A plays

$$\begin{array}{cc|ccc}
 & & q_1 & q_2 & q_3 \\
 & & b_1 & b_2 & b_3 \\
\hline
1-p & a_1 & 0 & -2 & 5 \\
p & a_2 & 4 & 10 & -1 \\
\hline
 & & 4p & 12p-2 & 5-6p
\end{array}$$



A bigger example

$$\begin{array}{ccc} & & \text{B plays} & q_1 & q_2 & q_3 \\ \text{A plays} & & & b_1 & b_2 & b_3 \\ 1-p & a_1 & & 0 & -2 & 5 \\ p & a_2 & & 4 & 10 & -1 \\ & & & 4p & 12p-2 & 5-6p \end{array}$$

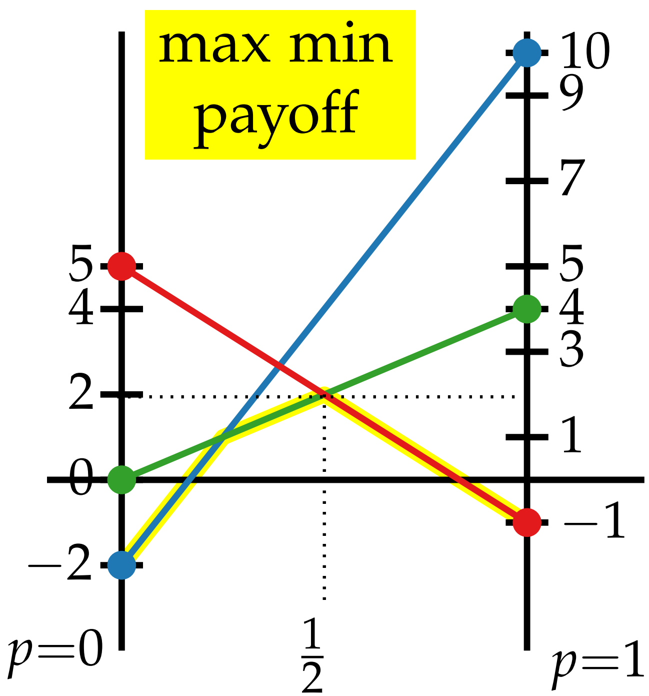


A bigger example

$$\begin{array}{cc|ccc} & & q_1 & q_2 & q_3 \\ & & b_1 & b_2 & b_3 \\ 1-p & a_1 & 0 & -2 & 5 \\ p & a_2 & 4 & 10 & -1 \\ & & 4p & 12p-2 & 5-6p \end{array}$$

B plays

A plays



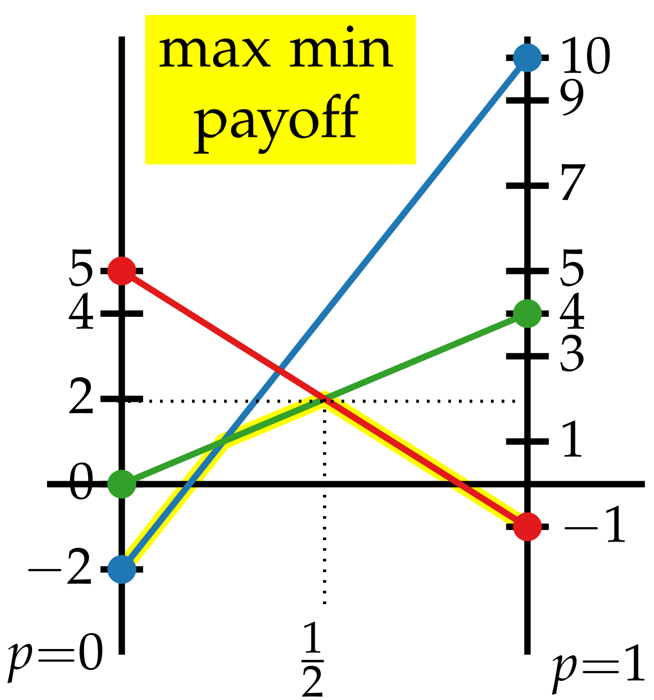
Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

A bigger example

B plays

A plays

$$\begin{array}{rrc}
 & & \begin{matrix} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{matrix} \\
\begin{matrix} 1-p \\ p \end{matrix} & \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} \\
 & & \begin{matrix} 4p & 12p-2 & 5-6p \end{matrix}
\end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

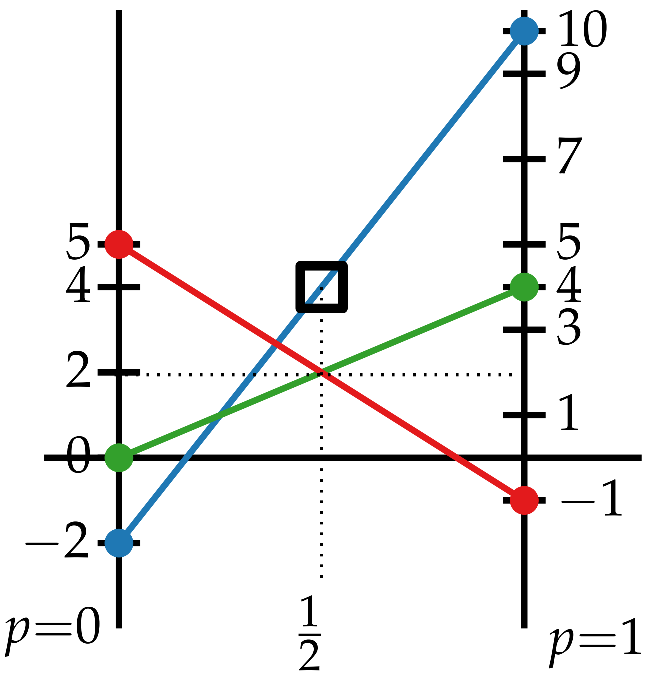
But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ ...

A bigger example

B plays

A plays

$$\begin{array}{ccc}
 & & \begin{array}{ccc} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{array} \\
\begin{array}{c} 1-p \\ p \end{array} & \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} \\
 & & \begin{array}{ccc} 4p & 12p-2 & 5-6p \end{array}
\end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ ...

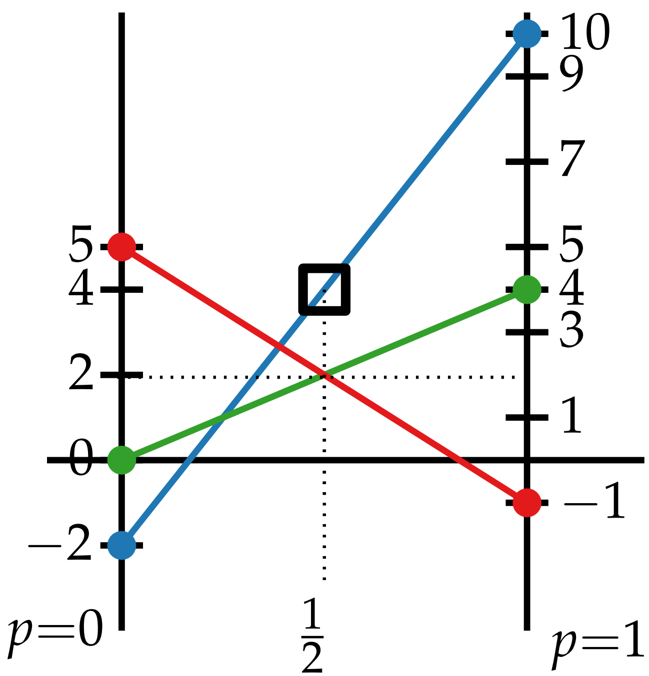
$\Rightarrow q_2 = 0$

A bigger example

B plays

A plays

$$
\begin{array}{cc|ccc}
 & & q_1 & q_2 & q_3 \\
 & & b_1 & b_2 & b_3 \\
\hline
1-p & a_1 & 0 & -2 & 5 \\
p & a_2 & 4 & 10 & -1 \\
\hline
 & & 4p & 12p-2 & 5-6p
\end{array}
$$



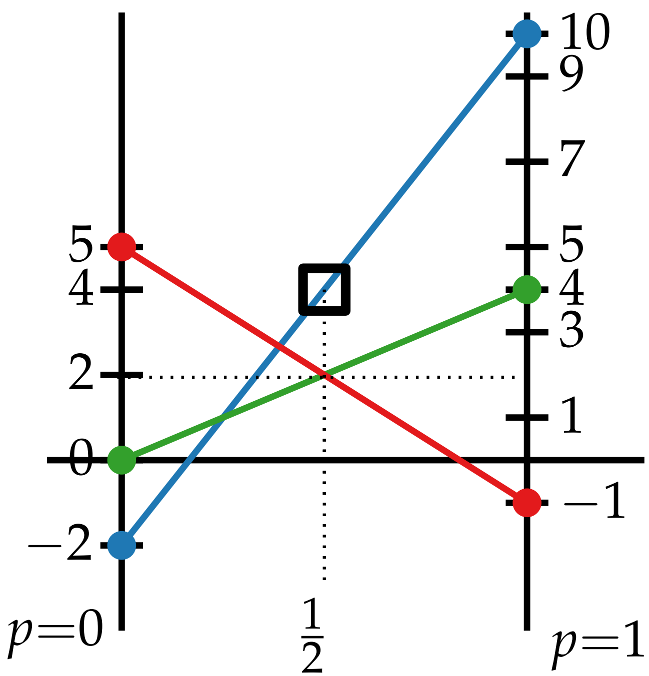
Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ ...

$\Rightarrow q_2 = 0$

A bigger example

$$
\begin{array}{lll|ccc|l}
 & & \text{B plays} & {\color{blue}q_1} & {\color{blue}q_2} & {\color{blue}q_3} & \\
\text{A plays} & & & b_1 & b_2 & b_3 & \\
{\color{blue}1-p} & a_1 & & 0 & -2 & 5 & -2q_2 + 5q_3 \\
{\color{blue}p} & a_2 & & 4 & 10 & -1 & \\
 & & & 4p & 12p-2 & 5-6p &
\end{array}
$$



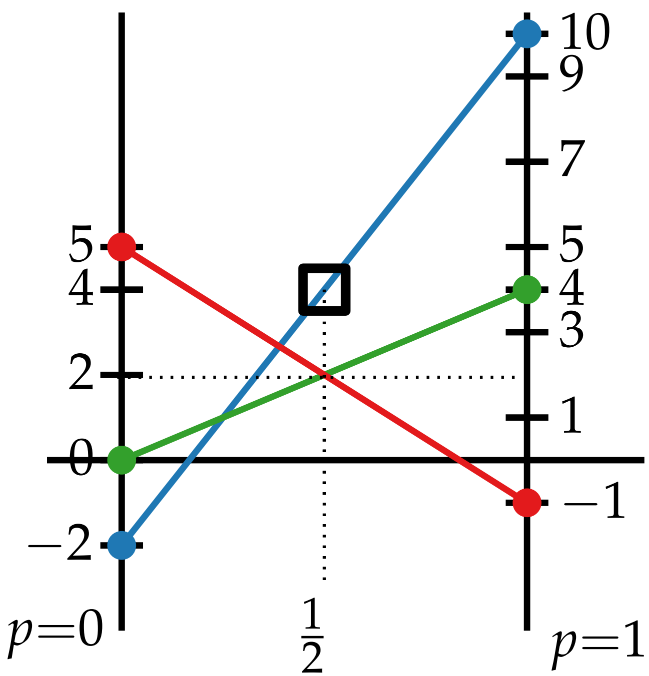
Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ . . .

$\Rightarrow q_2 = 0$

A bigger example

$$
\begin{array}{ccc}
 & \text{B plays} & \\
\text{A plays} & &
\end{array}
\qquad
\begin{array}{cc|ccc|c}
 & & q_1 & q_2 & q_3 & \\
 & & b_1 & b_2 & b_3 & \\
\hline
1-p & a_1 & 0 & -2 & 5 & -2q_2+5q_3 \\
p & a_2 & 4 & 10 & -1 & 4q_1+10q_2-1q_3 \\
\hline
 & & 4p & 12p-2 & 5-6p &
\end{array}
$$



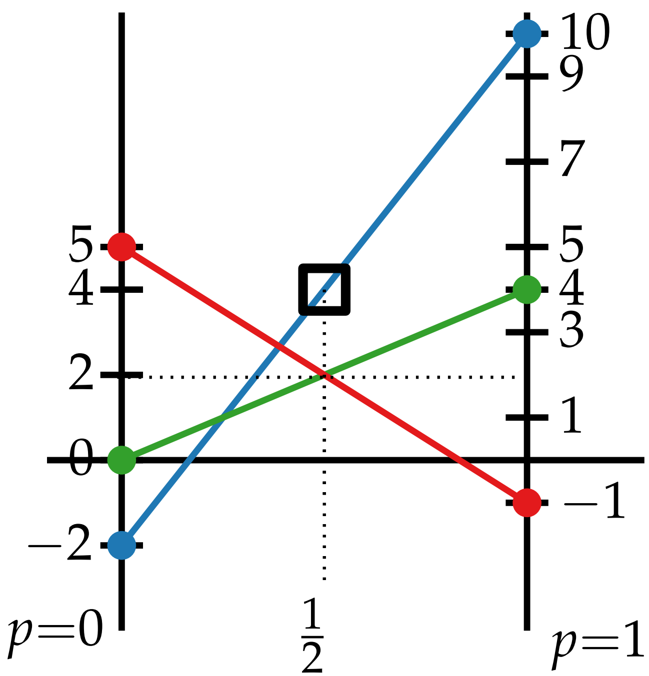
Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ . . .

$\Rightarrow q_2 = 0$

A bigger example

$$
\begin{array}{ccc|ccc|c}
 & \text{B plays} & & q_1 & q_2 & q_3 & \\
 \text{A plays} & & & b_1 & b_2 & b_3 & \\
1-p & a_1 & & 0 & -2 & 5 & -2q_2+5q_3 \\
p & a_2 & & 4 & 10 & -1 & 4q_1+10q_2-1q_3 \\
 & & & 4p & 12p-2 & 5-6p &
\end{array}
$$



Thus, $p=\frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q}=(q_1,q_2,q_3)$ . . .

$\Rightarrow q_2=0$

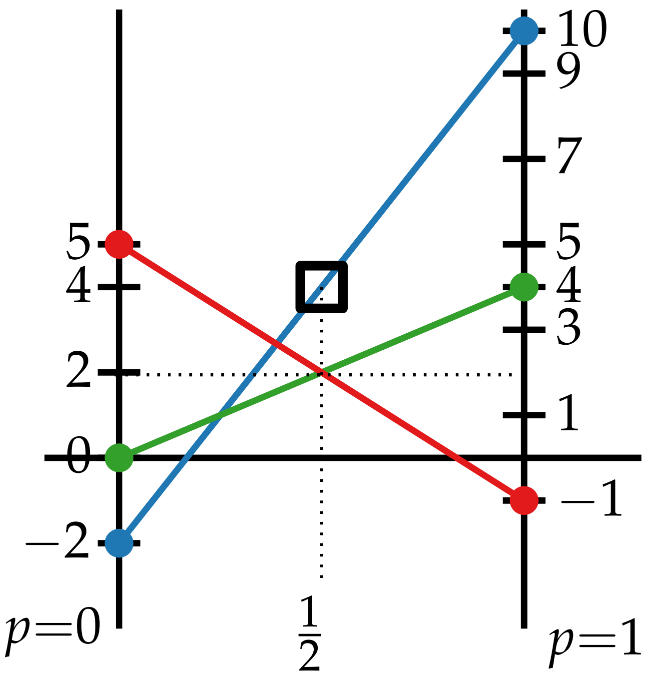
[Minimax thm. $\mathbf{p}S\mathbf{q}^T=v^\star$]

$(1-p)(-2q_2+5q_3)+p(4q_1+10q_2-1q_3)=v^\star$

$q_1(4p)+q_2(12p-2)+q_3(5-6p)=v^\star$

A bigger example

$$\begin{array}{cc|ccc|l} & \text{B plays} & q_1 & q_2 & q_3 & \\ \text{A plays} & & b_1 & b_2 & b_3 & \\ 1-p & a_1 & 0 & -2 & 5 & -2q_2+5q_3 \\ p & a_2 & 4 & 10 & -1 & 4q_1+10q_2-1q_3 \\ & & 4p & 12p-2 & 5-6p & \end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ …

$\Rightarrow q_2 = 0$

[Minimax thm. $\mathbf{p}S\mathbf{q}^T = v^\star$]

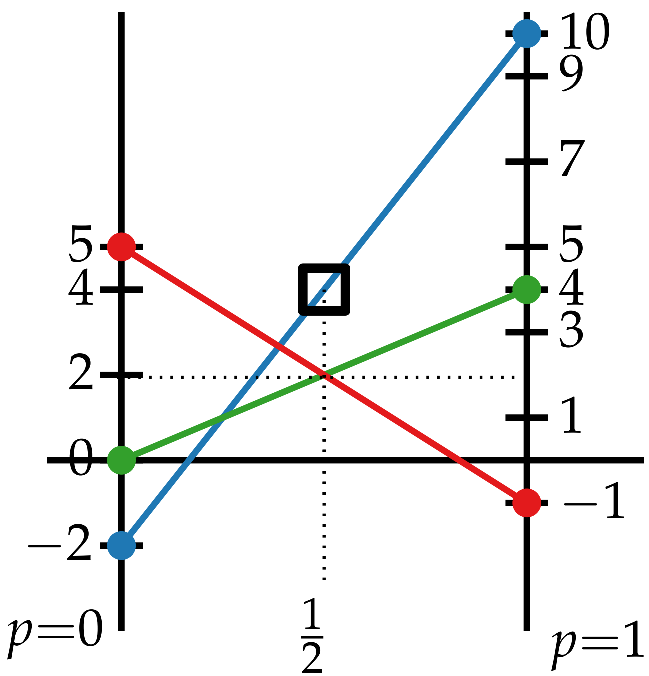
$(1-p)(-2q_2 + 5q_3) + p(4q_1+10q_2-1q_3) = v^\star$

$q_1(4p) + q_2(12p-2) + q_3(5-6p) = v^\star$

$p = 0 \rightsquigarrow -2q_2+5q_3 = 2$

A bigger example

$$\begin{array}{rrcl}
 & \text{B plays} & \begin{matrix} q_1 & q_2 & q_3 \end{matrix} & \\
\text{A plays} & & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} & \\
\begin{matrix} 1-p \\ p \end{matrix} & \begin{matrix} a_1 \\ a_2 \end{matrix} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} & \begin{matrix} -2q_2 + 5q_3 \\ 4q_1 + 10q_2 - 1q_3 \end{matrix} \\
 & & \begin{matrix} 4p & 12p-2 & 5-6p \end{matrix} &
\end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3) \ldots$

$\Rightarrow q_2 = 0$

[Minimax thm. $\mathbf{p}S\mathbf{q}^T = v^\star$]

$(1-p)(-2q_2 + 5q_3) + p(4q_1 + 10q_2 - 1q_3) = v^\star$

$q_1(4p) + q_2(12p - 2) + q_3(5 - 6p) = v^\star$

$p = 0 \rightsquigarrow -2q_2 + 5q_3 = 2$

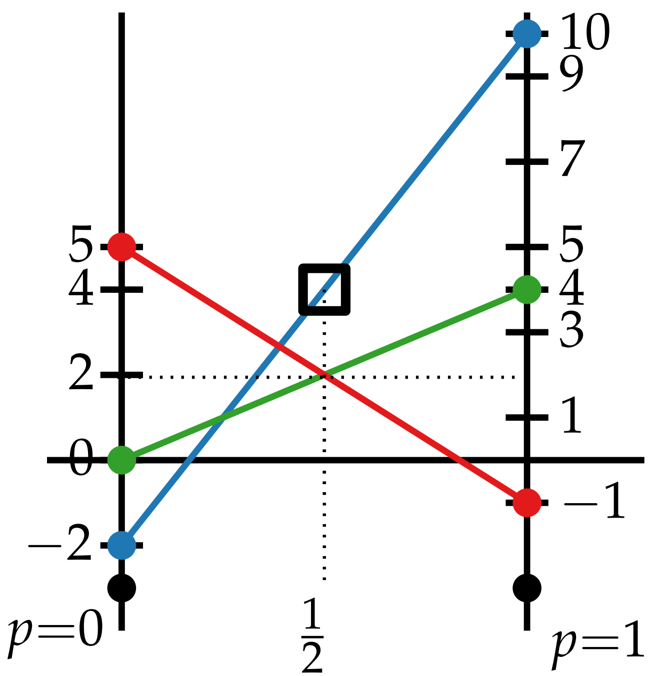
$p = 1 \rightsquigarrow 4q_1 - q_3 = 2$

A bigger example

B plays

A plays

$$\begin{array}{cccc}
 & & \begin{array}{ccc} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{array} & \\
\begin{array}{c} 1-p \\ p \end{array} & \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} & \begin{array}{c} -2q_2+5q_3 \\ 4q_1+10q_2-1q_3 \end{array} \\
 & & \begin{array}{ccc} 4p & 12p-2 & 5-6p \end{array} &
\end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ ...

$\Rightarrow q_2 = 0$

[Minimax thm. $\mathbf{p}S\mathbf{q}^T = v^\star$]

$(1-p)(-2q_2 + 5q_3) + p(4q_1+10q_2-1q_3) = v^\star$

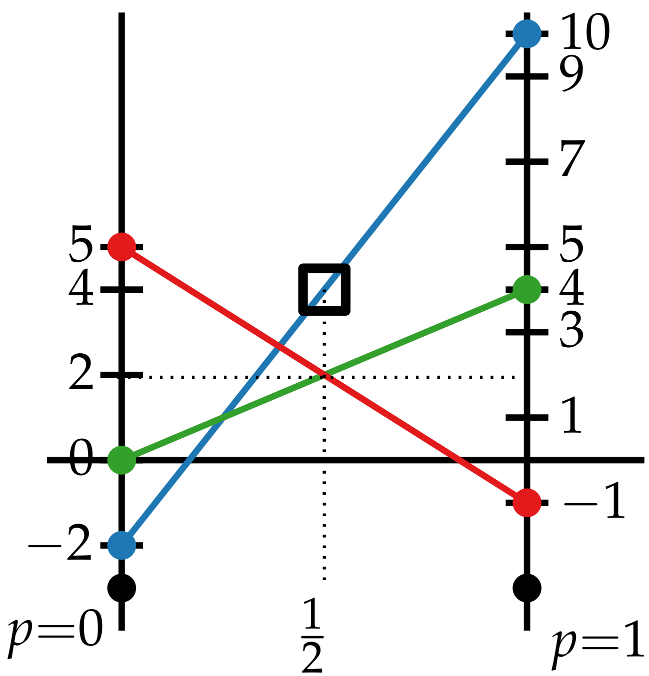
$q_1(4p) + q_2(12p-2) + q_3(5-6p) = v^\star$

$p = 0 \rightsquigarrow -2q_2+5q_3 = 2 \qquad \Rightarrow q_3 = \frac{2}{5}$

$p = 1 \rightsquigarrow 4q_1 - q_3 = 2 \qquad \Rightarrow q_1 = \frac{3}{5}$

A bigger example

$$\begin{array}{ccc}
 & \text{B plays} & \begin{array}{ccc} q_1 & q_2 & q_3 \\ b_1 & b_2 & b_3 \end{array} & \\
\text{A plays} & & & \\
\begin{array}{c} 1-p \\ p \end{array} & \begin{array}{c} a_1 \\ a_2 \end{array} & \begin{bmatrix} 0 & -2 & 5 \\ 4 & 10 & -1 \end{bmatrix} & \begin{array}{l} -2q_2+5q_3 \\ 4q_1+10q_2-1q_3 \end{array} \\
 & & \begin{array}{ccc} 4p & 12p-2 & 5-6p \end{array} &
\end{array}$$



Thus, $p = \frac{1}{2}$, and the **value** $v^\star$ of the game is 2.

But, we didn't find $\mathbf{q} = (q_1, q_2, q_3)$ ...

$\Rightarrow q_2 = 0$

[Minimax thm. $\mathbf{p} S \mathbf{q}^T = v^\star$]

$(1-p)(-2q_2 + 5q_3) + p(4q_1 + 10q_2 - 1q_3) = v^\star$

$q_1(4p) + q_2(12p-2) + q_3(5-6p) = v^\star$

$p = 0 \rightsquigarrow -2q_2 + 5q_3 = 2 \qquad \Rightarrow q_3 = \frac{2}{5}$

$p = 1 \rightsquigarrow 4q_1 - q_3 = 2 \qquad \Rightarrow q_1 = \frac{3}{5}$

$$q = \left(\tfrac{3}{5}, 0, \tfrac{2}{5}\right)$$

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

$$\begin{array}{c|cc} \text{A plays} \backslash \text{B plays} & b_1 & b_2 \\ \hline a_1 & & \\ a_2 & & \\ a_3 & & \end{array}$$

$$\begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array} \begin{array}{c} \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

- Is this the same game?

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

B plays

A plays

$$\begin{array}{c c} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

- Is this the same game?
  No, payoff/loss is reversed!

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

$$\begin{array}{c|c} \text{A plays} \setminus \text{B plays} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

- Is this the same game?
  No, payoff/loss is reversed!

$$\begin{array}{c|c} \text{A plays} \setminus \text{B plays} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & -4 \\ 2 & -10 \\ -5 & 1 \end{bmatrix} \end{array} = -S^T$$

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

$$\begin{array}{c} \text{A plays} \backslash \text{B plays} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} \end{array} \begin{array}{c} \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

- Is this the same game?
  No, payoff/loss is reversed!

$$\begin{array}{c} \text{A plays} \backslash \text{B plays} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} \end{array} \begin{array}{c} \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{bmatrix} 0 & -4 \\ 2 & -10 \\ -5 & 1 \end{bmatrix} \end{array} = -S^T$$

- What about this game?

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

$$\begin{array}{cc} & \text{B plays} \\ \text{A plays} & \begin{array}{c c} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} = S^T \end{array}$$

- Is this the same game?
  No, payoff/loss is reversed!

$$\begin{array}{cc} & \text{B plays} \\ \text{A plays} & \begin{array}{c c} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & -4 \\ 2 & -10 \\ -5 & 1 \end{bmatrix} = -S^T \end{array}$$

- What about this game?
  This is the same game as S.

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

$$\begin{array}{c c} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

B plays / A plays

- Is this the same game?
  No, payoff/loss is reversed!

$$\begin{array}{c c} & \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{array}{c} a_1 \\ a_2 \\ a_3 \end{array} & \begin{bmatrix} 0 & -4 \\ 2 & -10 \\ -5 & 1 \end{bmatrix} \end{array} = -S^T$$

B plays / A plays

- What about this game?
  This is the same game as S.

Flip the Matrix? [Let S be the (simplified) game of prev. slide]

B plays / A plays

$$\begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array}\begin{array}{c} \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{bmatrix} 0 & 4 \\ -2 & 10 \\ 5 & -1 \end{bmatrix} \end{array} = S^T$$

- Is this the same game?
  No, payoff/loss is reversed!

B plays / A plays

$$\begin{array}{c} \\ a_1 \\ a_2 \\ a_3 \end{array}\begin{array}{c} \begin{array}{cc} b_1 & b_2 \end{array} \\ \begin{bmatrix} 0 & -4 \\ 2 & -10 \\ -5 & 1 \end{bmatrix} \end{array} = -S^T$$

- What about this game?
  This is the same game as S.